

The Central Nervous system: Anatomy and Physiology

Integrated Biomedical and Clinical Sciences OMFS 512

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Nervous system

86 billion neurons in
the human brain

1000 trillion synapsis

Milky way

200-400 billion stars

May 20, 2013 | By: Bradley Voytek

Are There Really as Many Neurons in the Human Brain as Stars in the Milky Way?

Aa Aa

This is a phrase a lot of science communicators like to use because giving people a sense of scale when it comes to large numbers is so difficult why journalists report distances as number of football fields, mass as number of fully-loaded 747s, energy in terms of Hiroshima bombs, etc.

Even though we can't conceive of the number of stars in the Milky Way or the number of neurons in the human brain, equating the two gives pec sense of enormity. And as conscious beings we like to find patterns, and we find equivalencies interesting, especially when the things being equ: "important" or "epic" (like neurons and stars).



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Learning Objectives

Learning Outcomes

1. Describe the structures of the nervous system.
2. Describe the structure of the brain.
3. Describe the function of different brain areas.
4. Describe the function of the brain ventricles and define hydrocephalus.

Outline

- Components of the nervous system
- Embryonic development of the nervous system.
- Histology of the nervous system.
- Anatomy and Physiology of the brain.
 - Telencephalon
 - Mesencephalon
 - Rhombencephalon
- Ventricular system and hydrocephalus

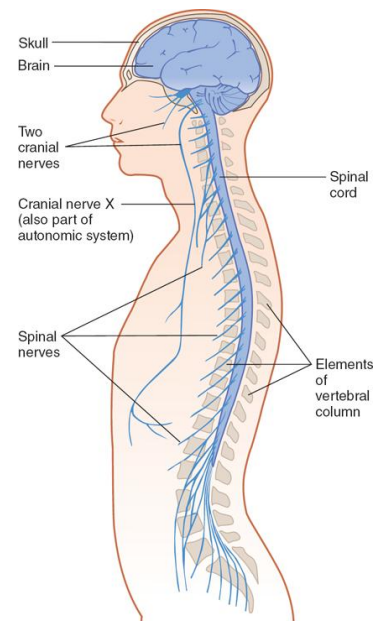
Nervous System

Central Nervous System (CNS)

Brain and spinal cord shielded by the meninges and CSF-filled space and enclosed in bone (skull and vertebral column).

Peripheral nervous system (PNS)

Formed by ganglia, cranial and spinal nerves. Organized into an autonomic and somatic



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Source: Stephen G. Waxman: *Clinical Neuroanatomy*, 30th Edition
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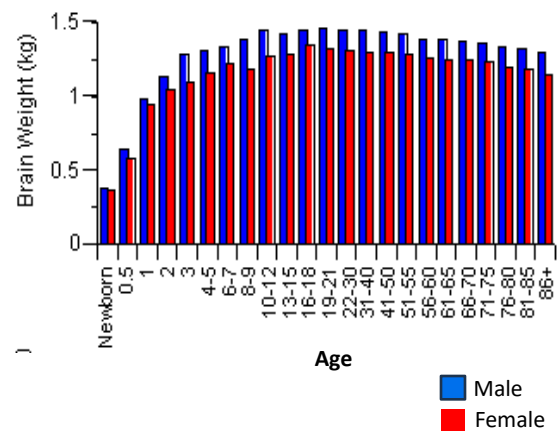
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Functional organization

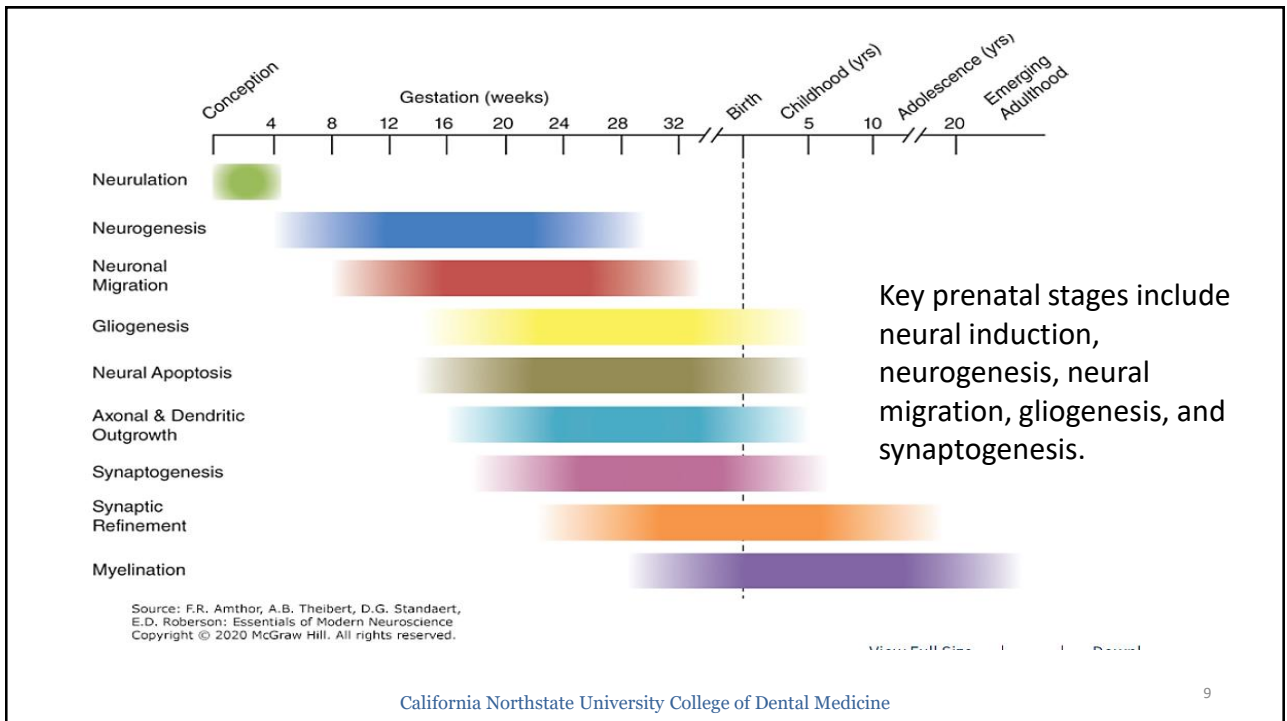
- Neurons (nerve cells), interneurons & glial cells.
- Nuclei= cluster of neural cell bodies in CNS. Ganglia = group of neural cell bodies with common function outside the CNS.
- Synapses = area where nerve cells convey signals to one another. Neurotransmitters play an imp. role in relaying information.
- Tracts and Commissures:
 - Tracts are nerve fiber bundles that connect groups of neurons in vertical manner. Example: ascending and descending tracts.
 - Commissures: Horizontal (lateral) nerve fiber connections in the brain, like corpus callosum.

Embryonic development

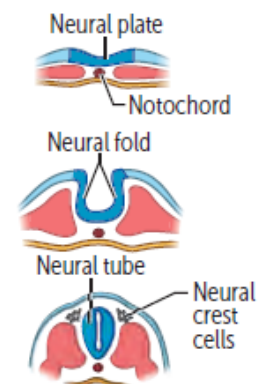
- Development of the human nervous system involves the generation of the CNS and PNS during embryonic, fetal, and postnatal periods.
- At birth, the human brain is approximately 350 to 400 g, and it grows till 1300 to 1400 g in adults.



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- Around the 3rd week of gestation, the nervous system start developing from the ectoderm layer.
- The ectoderm thickens to form neuroectoderm that generates the neural plate (Sonic Hedgehog protein).
- The neural plate folds inwards to create a neural tube and neural crest.
 - The anterior end of the neural tube forms the brain structures while the posterior develops into the spinal cord. Neural tube cells give rise to neurons and glia in the CNS.
- Neural crest cells migrate and generate the neurons and glia of PNS and other cell types like melanocytes, odontoblasts and chromaffin cells.



Errors in development of the nervous system from exposure to teratogens or genetic anomalies & others lead to congenital or sensory, motor or behavioral deficits.

Neural tube defects Congenital malformation due to genetic factors, increased alcohol intake during pregnancy, and exposure to certain drugs (like valproic acid, anticonvulsant) or folic acid deficiency. Neural tube defects include anencephaly and spina bifida.

- **Anencephaly:** severe neural tube defect. Failure of closure of the anterior neural tube. Characterized by failure of brain development and missing skull bones as well as forebrain. Infants die shortly after birth.



Source: Kemp WL, Burns DK, Brown TG: *Pathology: The Big Picture*:
www.accessmedicine.com

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- ***Spina bifida***: Characterized by defective closure of posterior neuropore. Several subtypes according to severity.
 - Spina bifida occulta is a mild and often asymptomatic form. One or more vertebrae spine don't fully close, but the spinal cord is undamaged.
 - Severe cases: protrusion or herniation of the spinal cord and meninges through the unfused areas of vertebral column.



Blood Supply to the Brain

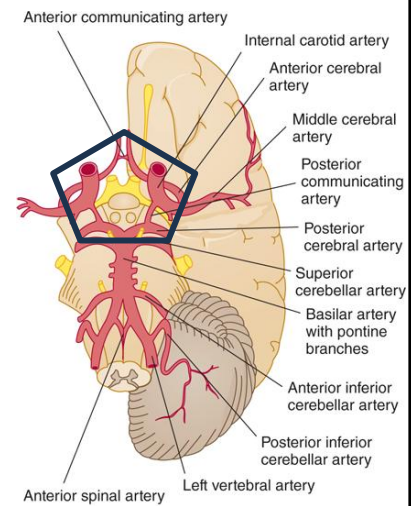
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Brain Blood Supply

The brain and spinal cord depend on the continuous supply of oxygenated blood. Blood flow interruptions is a major risk for strokes. Loss of consciousness can occur in less than 15 sec.

The brain receives around 15 % of daily the cardiac output with an anterior circulation and posterior circulation. At the central cranial base, the anterior and posterior circulation connect to form circle of Willis.



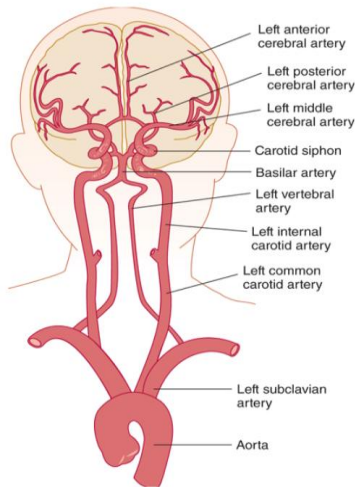
Source: Stephen G. Waxman: *Clinical Neuroanatomy, 30th Edition*
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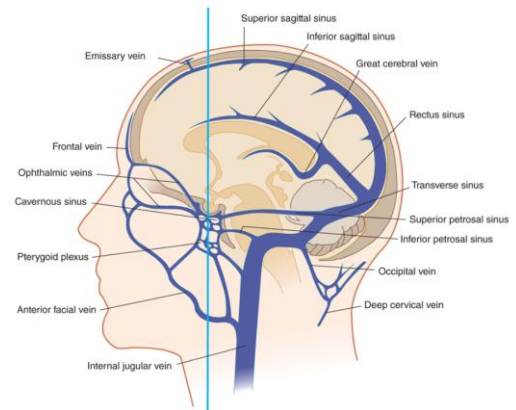
- **Arterial supply**

- Anterior: pair of anterior carotid arteries
- Posterior: pair of vertebral arteries.



- **Venous drainage** through the internal Jugular vein, from:

- Venous sinuses.
- Network of cerebral veins



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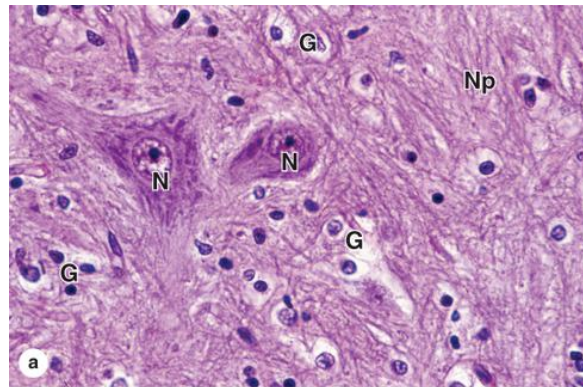
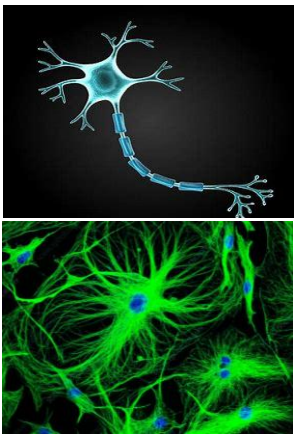


Cellular Neuroscience

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Histology: A network **neurons** supported by **glial cells**



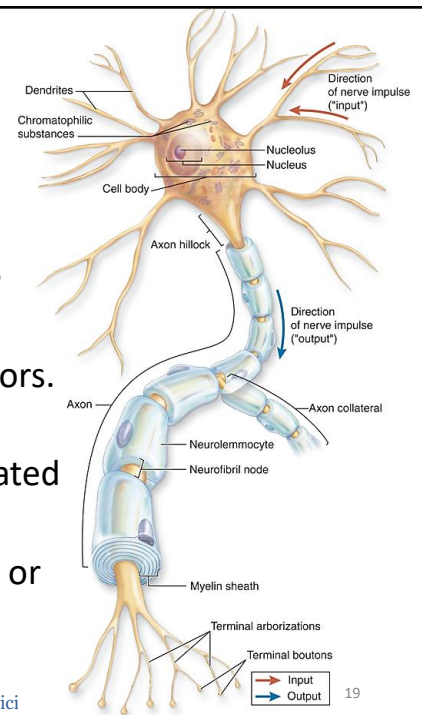
Source: Anthony L. Mescher:
Junqueira's Basic Histology Text and Atlas, 16e
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Neurons

- Cell body that contains the nucleus and other organelles, so it can be considered the genetic and metabolic center of the neuron. Cytoplasm include lots of Nissl bodies.
- Dendrites receive stimuli or input via chemoreceptors.
- Axon carry the nerve impulses away from the cell body towards axon terminal (send output), myelinated or non myelinated.
- Axon terminal form synapses with the next neuron or muscle.



Citation: CHAPTER 19 THE NERVOUS SYSTEM, Janson LW, Tischler ME. *The Big Picture: Medical Biochemistry*, 2018. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=2355§ionid=185845454> Accessed: February 06, 2022

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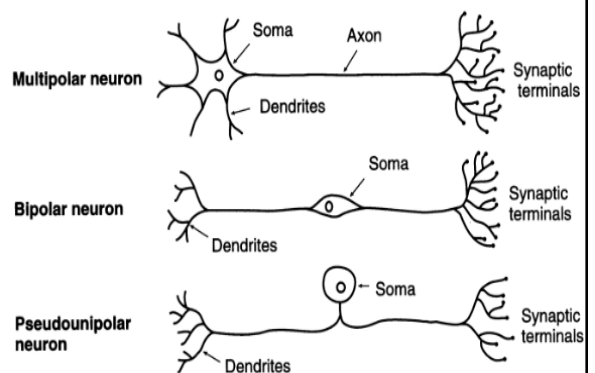
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According to the number of processes:

a) Multipolar: most common, multiple dendrites originating from the cell soma and single axon, e.g: CNS, somatic NS

b) Bipolar: two processes that extend from opposite sides of a round or oval-shaped cell body. Limited locations, e.g: sensory neurons in retina

c) Pseudounipolar: a single process that bifurcates shortly after leaving the soma, e.g sensory neurons of PNS



Types of nerve fibers

	General Fiber Type and Example	Sensory Fiber Type and Example	Diameter	Conduction Velocity
Myelinated	A-alpha Large α -motoneurons	Ia Muscle spindle afferents	Largest	Fastest
		Ib Golgi tendon organs	Largest	Fastest
	A-beta Touch, pressure	II Secondary afferents of muscle spindles; touch and pressure	Medium	Medium
	A-gamma γ -Motoneurons to muscle spindles (intrafusal fibers)	—	Medium	Medium
	A-delta Touch, pressure, temperature, and pain	III Touch, pressure, fast pain, and temperature	Small	Medium
	B Preganglionic autonomic fibers	—	Small	Medium
	C Slow pain; postganglionic autonomic fibers	IV Pain and temperature (unmyelinated)	Smallest	Slowest
Unmyelinated				

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Functional organization of neurons

a. Sensory (afferent) neurons receive stimuli from the internal and external environments and conduct impulses towards the CNS for processing and analysis. Sensory neurons are responsible for “perception”.

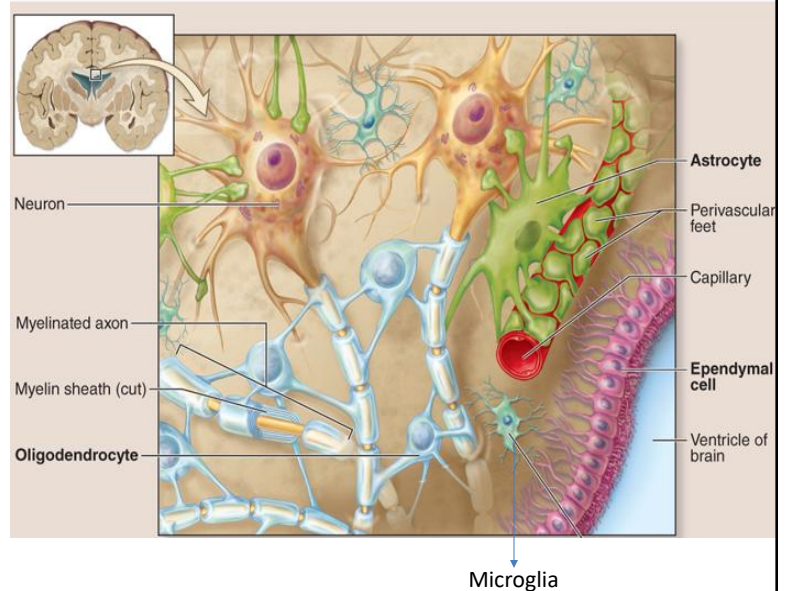
b. Motor (efferent) neurons conduct impulses from the CNS towards effector tissues (muscles, and glands). Responsible for “action”.

c. Interneurons (intercalated neurons). They can connect sensory and motor neurons and regulate the signals transmitted between neurons.

Glial population

Glial cells are more abundant than neurons (about 10 times). Function to support neuronal function and survival. Classified into microglia and macroglia.

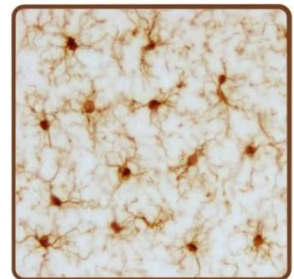
- CNS has oligodendrocytes, astrocytes, ependymal cells, and microglial cells.
- PNS has Schwann cells and Satellite cells.



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- 1) **Microglia:** immune cells of the CNS. Derived from monocytes (not a neural progenitor). Considered as a tissue-resident macrophage.
- They have branched processes that enable their motion between grey and white matter.
 - Immune defense mechanism. i.e, remove cellular debris resulting from injury, infection, and disease by phagocytosis.
 - Secrete immunoregulatory factors.

MICROGLIA**IMMUNOCYTOCHEMISTRY**

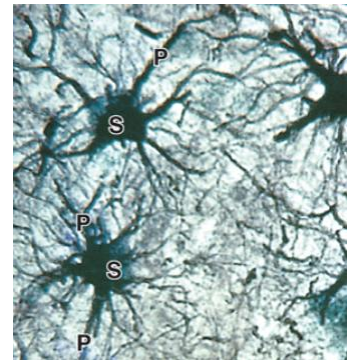
2) Macrogia: several types including astrocytes, oligodendrocytes, ependymal cells and Schwann cells.

a) Astrocytes:

Most common type of glial cell in the CNS. Produce structure protein GFAP (glial fibrillary acid protein).

Functions include:

- Scaffold to guide proper neural migration in brain development. Structural support for nervous tissue.
- Synaptic transmission, uptake and degrade some locally released neurotransmitters.
- Neural protection in injury and disease. Proliferate after CNS injury to fill in the tissue (forming astrocytic scar).
- Form the BBB through supporting the endothelial cells by astrocyte end feet.
- Transfer nutrients from the blood to the neurons.

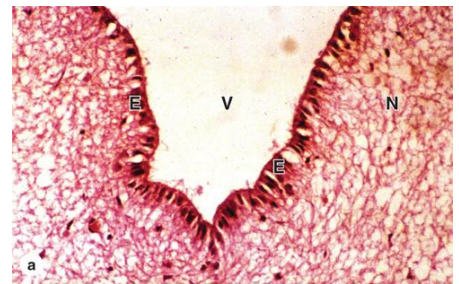


Star-shaped astrocytes. P= cytoplasmic processes; S= glial cell body (soma)

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b) Ependymal Cells

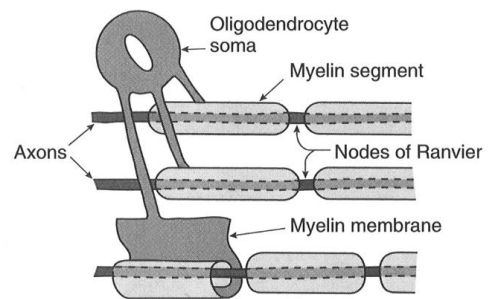
- Single-layer neuroepithelial lining for the ventricles of columnar shape .
- Functions to produce the cerebrospinal fluid.
- In some CNS locations, they have cilia to facilitate CSF movement or microvilli to help with CSF absorption.



E=ependymal cell; V= Ventricle

c) Oligodendrocytes

- Found only in the CNS.
- Oligodendrocyte have long processes that wrap around several neuronal axons.
- Function to produce myelin sheath for neural axons in the CNS.
- The **myelin sheath** electrically insulates the axon and facilitates rapid transmission of nerve impulses.



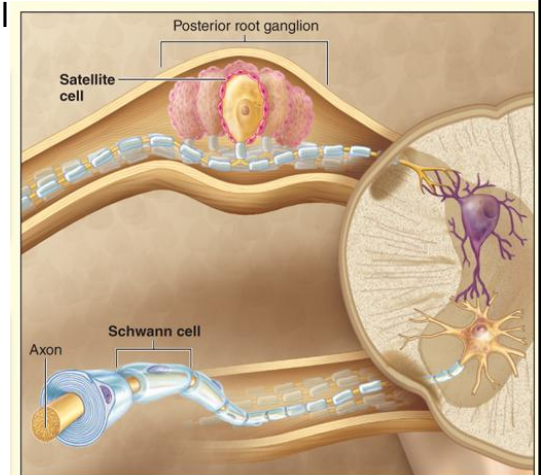
d) Schwann cells

- Found in the peripheral nervous system. Each Schwann cell is associated with one neuronal axon. Also named neurolemmocytes.
- Produce myelin sheath that insulates the axons, leading to faster transmission of nerve impulse.

e) Satellite Cells:

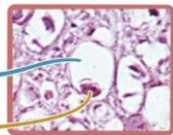
Found in PNS.

Functions for electricity insulation and nutrient/waste exchange along ganglia.



Macroscopically: Grey and White Matter

WHITE MATTER



WHITE MATTER

GREY MATTER

~ NEURON CELL BODIES

~ DENDRITES

~ ASTROCYTES

~ MICROGLIAL CELLS



SPINAL CORD

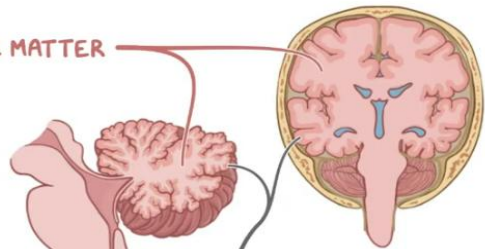
~ WHITE MATTER in the PERIPHERY

~ GREY MATTER CLOSER to the CENTER

↳ "H" or BUTTERFLY-SHAPED

GREY MATTER

~ OUTERMOST PORTION of the
CEREBRUM & CEREBELLUM

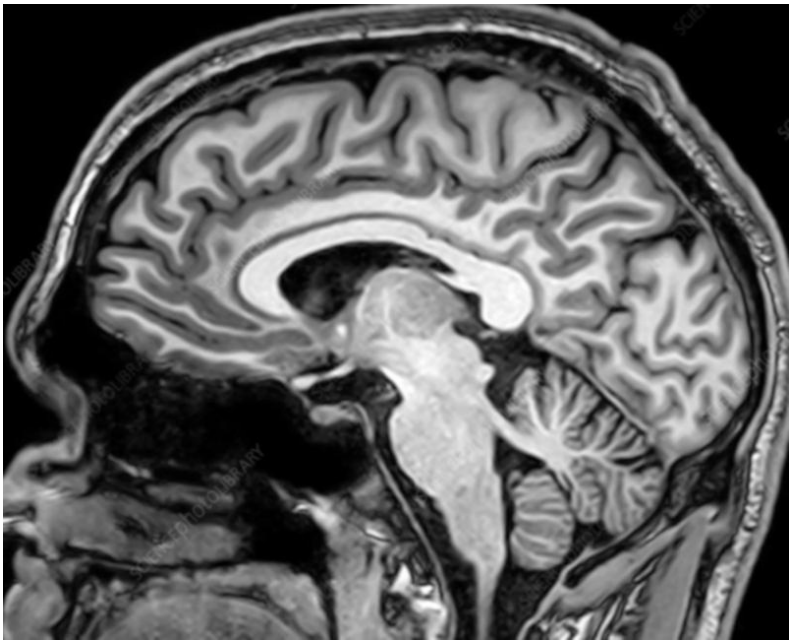


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Brain

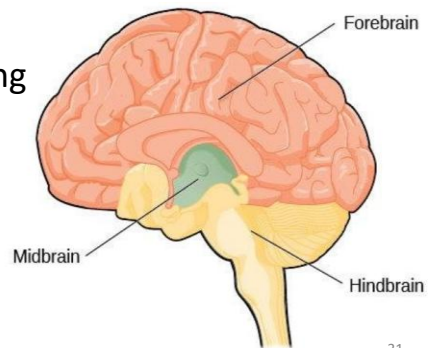
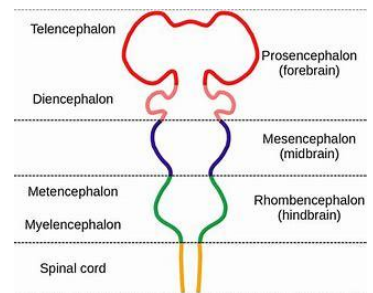


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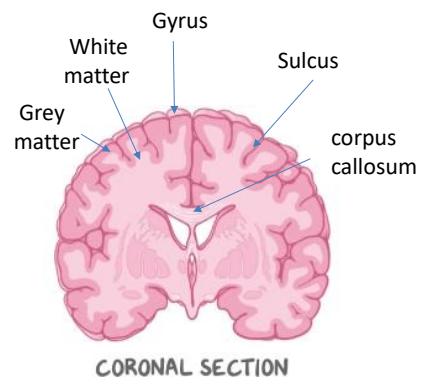
Anatomically, the brain is subdivided into:

- **The forebrain**, further subdivided into a telencephalon and diencephalon.
- **The midbrain**, lies at the rostral end of pons, and it contains the superior and inferior colliculi, cerebral peduncle and substantia nigra. The substantia nigra contains the dopamine producing neurons and it plays important function in movement control and reward system.
- **The hindbrain** consists of pons, medulla and the cerebellum.



Telencephalon “Cerebrum”

- Cerebrum is longitudinally divided into two right and left hemispheres connect by a bundle of nerve fibers (commissure) “corpus callosum”.
- Folding of cerebrum creates gyri and sulci. They increase the surface area and create boundaries between different brain regions.
- Composed of cortex, cerebral white matter and subcortical structures (like basal ganglia and hippocampus).



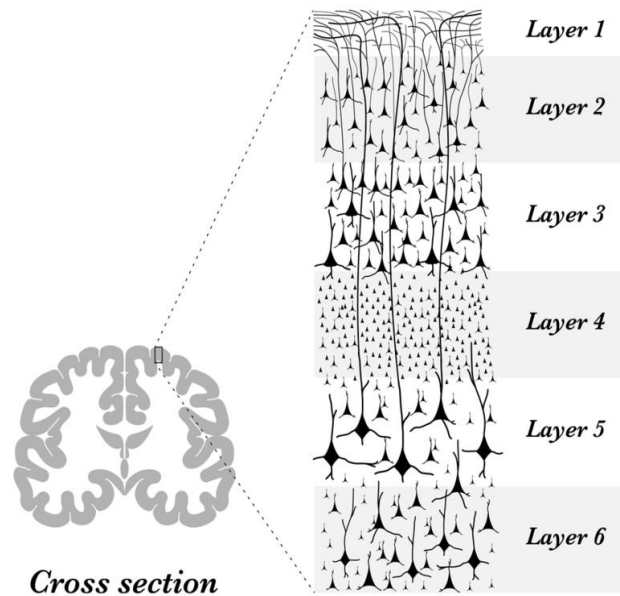
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A) Cerebral Cortex

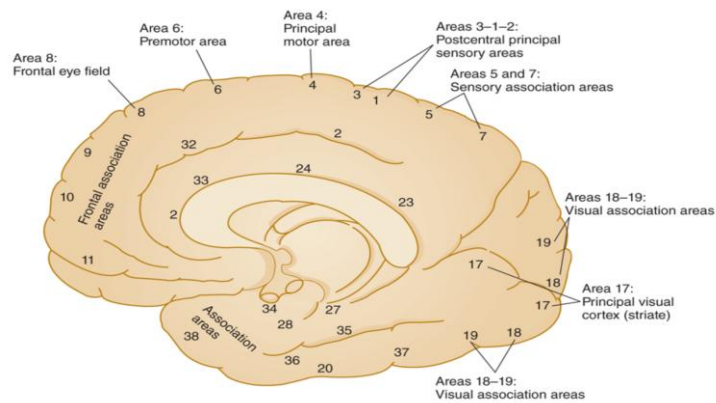
cytoarchitecture

- The outer layer of the cerebrum is termed “cerebral cortex”.
- Synchronized neural migration during embryonic developments produce six neural layers.
- Structural abnormalities of the cerebral cortex play important role in the etiology of several neurological disorders.



Cross section

Brodmann areas



Source: Stephen G. Waxman: *Clinical Neuroanatomy*, 29e
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The Brodmann classification classified the cortex into 52 regions based on the neural shapes and neural arrangement. This classification aided in mapping various brain functions to their specific cortical areas.

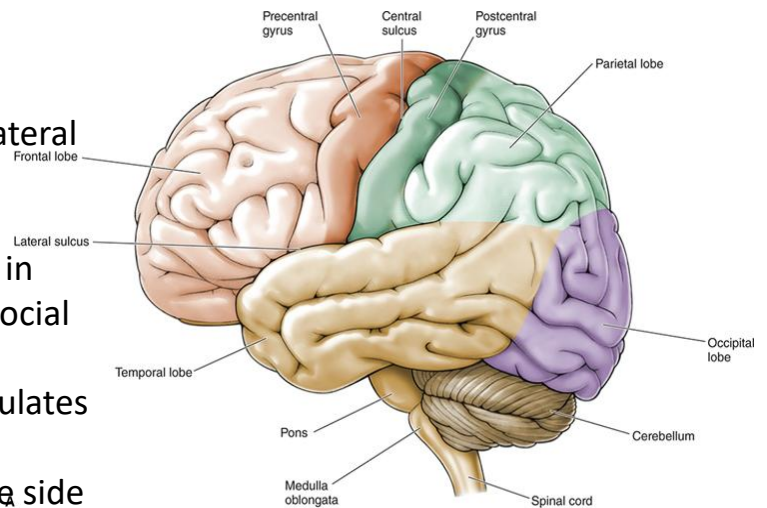
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cerebral cortex is divided into functional lobes by the action of major fissures and sulci:

The frontal lobe:

- Largest lobe in humans.
- Bound by central sulcus and lateral sulcus.
- Includes:
 - Prefrontal cortex, involved in judgment, reasoning and social behavior.
 - Primary motor cortex, regulates motor output (voluntary movement) to the opposite side of the body.

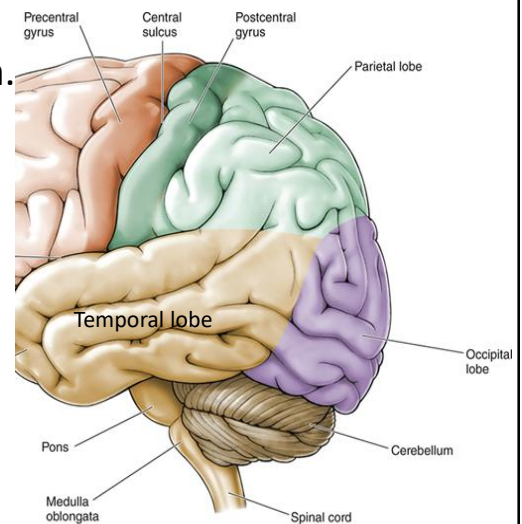


Citation: CHAPTER 16 Brain, Morton DA, Foreman K, Albertine KH. The Big Picture: Gross Anatomy, 2e, 2019.

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- **The parietal lobe** contains primary sensory cortex. Function: directs the integration of sensory input and attention.
- **Occipital lobe** located in the posterior cranial fossa, and it contains the primary visual cortex.
- **The temporal lobe** in the middle cranial fossa below the lateral sulcus. Functions include auditory processing, language recognition, and formation of new and long-term memories.

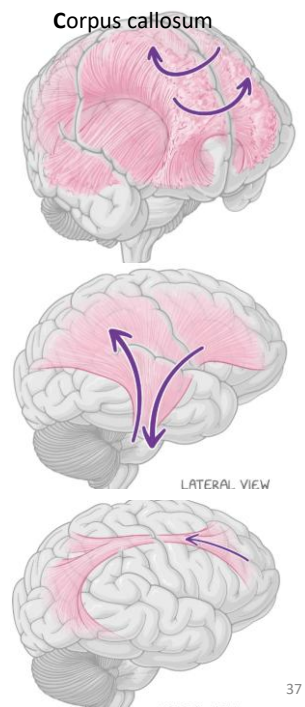


b) Cerebral White matter

A. Commissural fibers: Corpus callosum is a large bundle of fibers that crosses the longitudinal cerebral fissure and interconnects the left and right hemispheres. It contains millions of axons (myelinated and non-myelinated axons). It permits the two hemispheres to communicate together.

B. Projection fibers: connect the cerebral cortex with lower portions of the brain or the spinal cord. Include afferent and efferent fibers to enable the cortex of receiving or transmitting signals.

C. Association Fibers: connect various regions within the same hemisphere and permit the cortex to function as a coordinated unite.



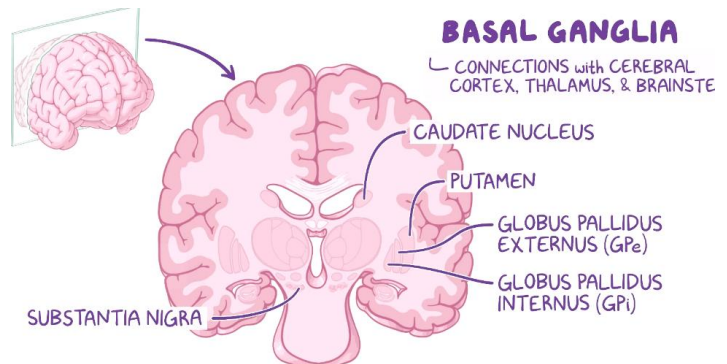
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c) Basal Ganglia

The basal ganglia are deep subcortical neurons composed of different structures. It consists of the striatum (caudate nucleus and putamen), globus pallidus and substantia nigra.

Responsible for thalamus-cortical neural signaling through connection with areas of the cerebral cortex, thalamus, and brainstem & others.



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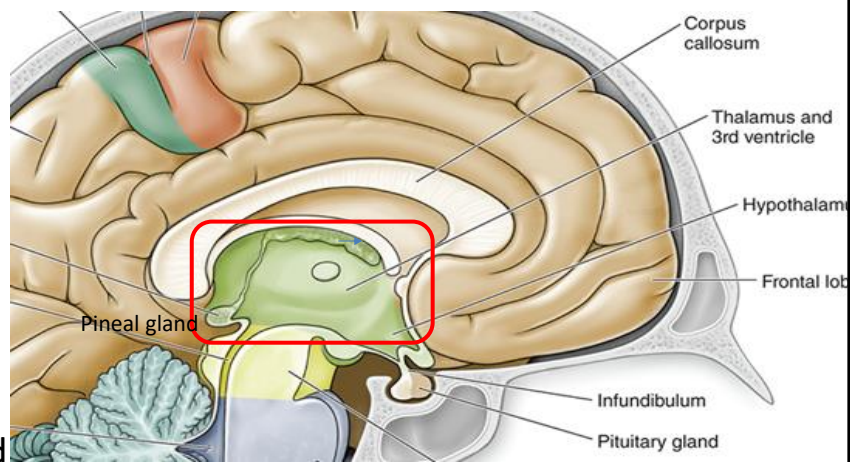
Function:

- Executing fine motor movement (can start and control movement and inhibit undesired movement).
 - Damage to substantia nigra is associated with pathologies like Parkinson's disease.
 - Lesions in the striatum occur in patients with Huntington disease.
- Play a role in cognition and decision-making.
- Motivation and brain reward system.

Diencephalon

The diencephalon is located between the cerebrum and the brainstem.

Composed of the thalamus and hypothalamus, pineal and posterior part of pituitary gland.



Citation: CHAPTER 16 Brain, Morton DA, Foreman K, Albertine KH. *The Big Picture: Gross Anatomy*, 2e, 2019.
Available at: <https://accessmedicine.mhmedical.com/content.aspx?sectionid=202020846&bookid=2478> Accessed: January 30, 2022

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a) Thalamus

Surrounds the third ventricle. Located in the both hemispheres.

Functions:

- Relay station that relay sensory and motor signals to the cerebral cortex.
- Regulate the cortical activation, memory and visual input.

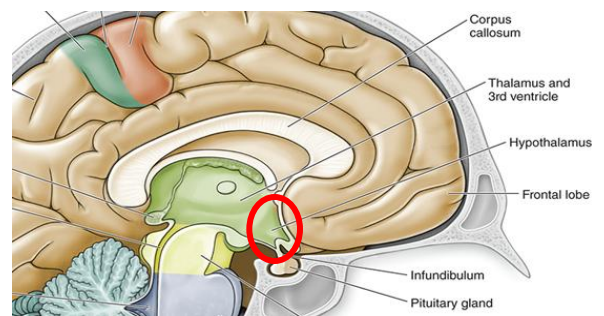
Include five sets of functional neural nuclei that relay different signals. For example, ventral posterolateral (VPL) nucleus relays pain sensation to the primary somatosensory cortex via the spinothalamic tract.

b) Hypothalamus

- Beneath the thalamus. Composed of 12 nuclei, each with specific function.

Functions

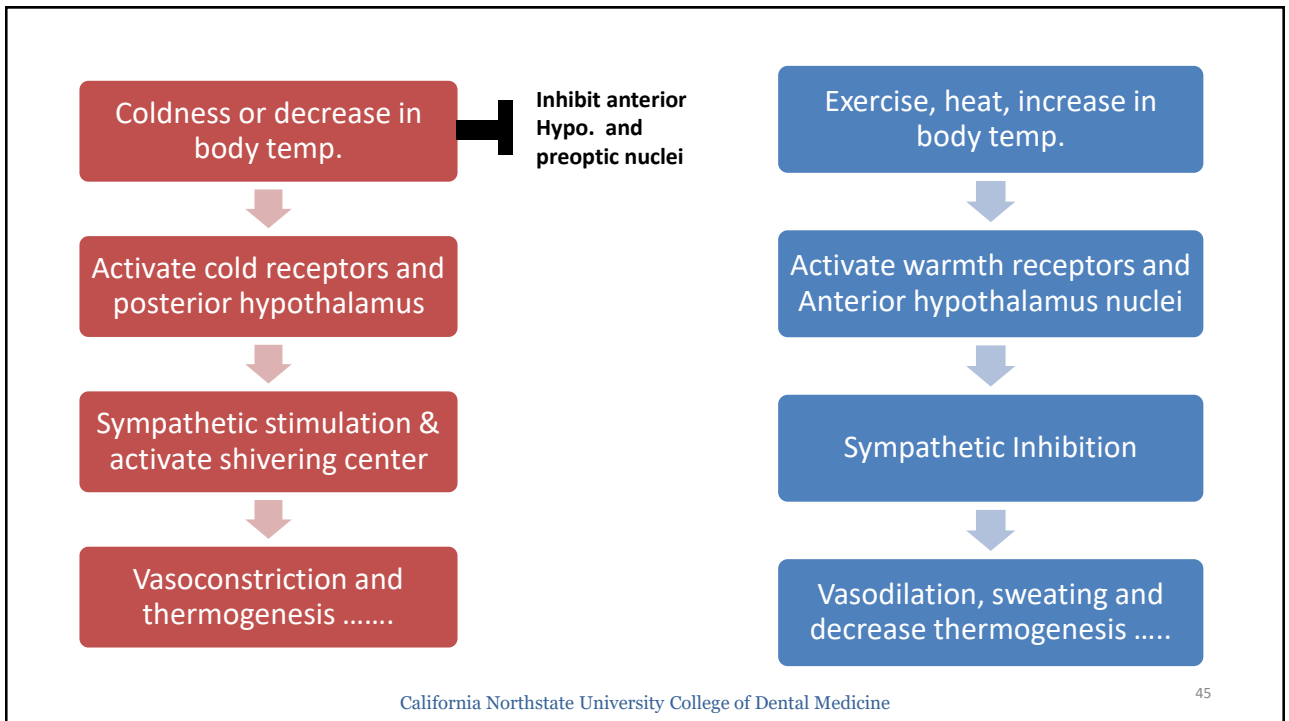
- Maintenance of homeostasis and emotional control
- Endocrine control through the hypothalamic-hypophyseal system that connects the hypothalamus with posterior pituitary.
- Functions in temperature regulation by inducing shivering to increase heat production or orchestrate heat-loss.
- Regulation of food intake.



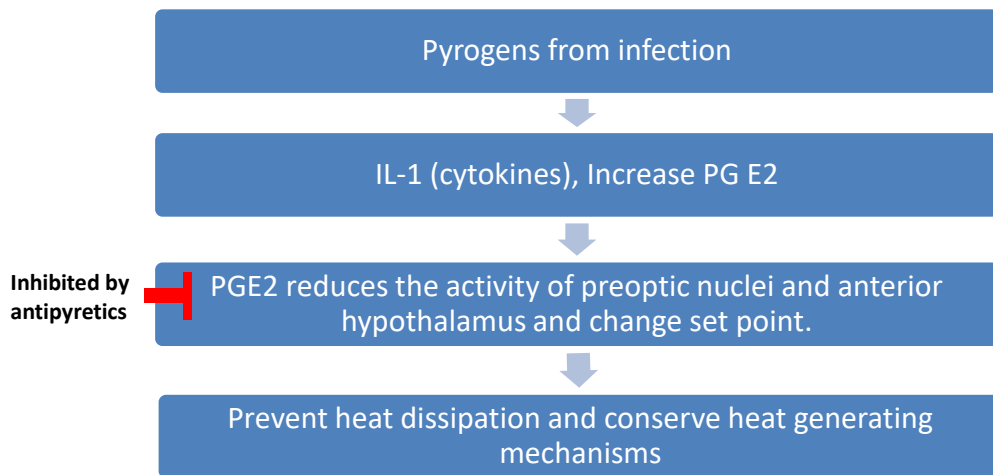
Regulation of Body Temperature

- Thermoreceptors provide temperature information to the preoptic area (in hypothalamus) to maintain a constant core body temperature.
- Body temperature is regulated via temperature-regulating centers in the hypothalamus.
 - **Heat production:** If the core temperature is below the hypothalamus set point, then the posterior hypothalamus will induce heat-generating mechanisms like
 - Increased metabolism (via thyroid gland) to increase heat production.
 - Shivering to increase rate of body heat production.
 - Inhibit sweating,
 - Promote vasoconstriction to diminish loss of body heat from the skin.

- Heat loss: If the core temperature is above the hypothalamus set point, the anterior hypothalamus will regulate mechanisms to enable heat loss like:
 - Vasodilation of skin blood vessels.
 - Sweating to enable heat loss via evaporation.
 - Decrease in heat production.

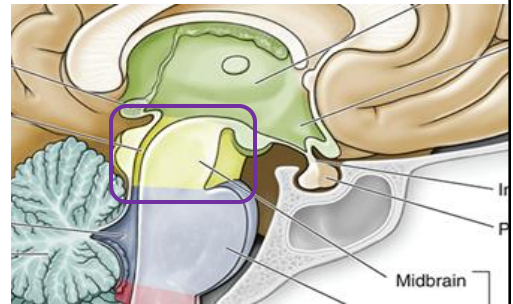


Fever or Pyrexia



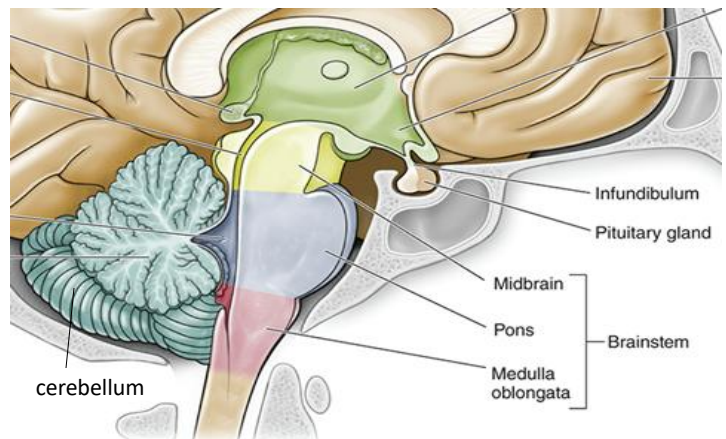
Midbrain (Mesencephalon)

- Narrow region that connects forebrain and hindbrain. It has nerve tracts superior and inferior colliculi, cerebral peduncle and **substantia nigra**.
- It contains granules filled **neuromelanin** pigment, making this region look darker.
- The substantia nigra contains the dopamine producing neurons. Plays important function in motor control and reward system. Connected to basal ganglia via neural tracts.



Hindbrain (Rhombencephalon)

- **Hindbrain** includes the pons, medulla oblongata and cerebellum.
- **Brainstem** consists of the midbrain, and pons, and medulla oblongata. Injury lead to neurological & threatening consequences.



an, Kurt H. Albertine: *The Big Picture: Gross Anatomy*, 20
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- **The pons** relay sensory information between cerebrum and spinal cord. Contains nerve nuclei for sensation, eye movement and auditory functions. Regulate arousal/sleep cycle.
- **The medulla** serves as reflex center with cardiovascular and respiratory control. Origin of several cranial nerves. Contain ascending and descending fiber tracts that connects the brain stem to the spinal cord. Responsible for the vomiting reflex.

Neural tracts from
and to the
telencephalon
passes through the
brain stem

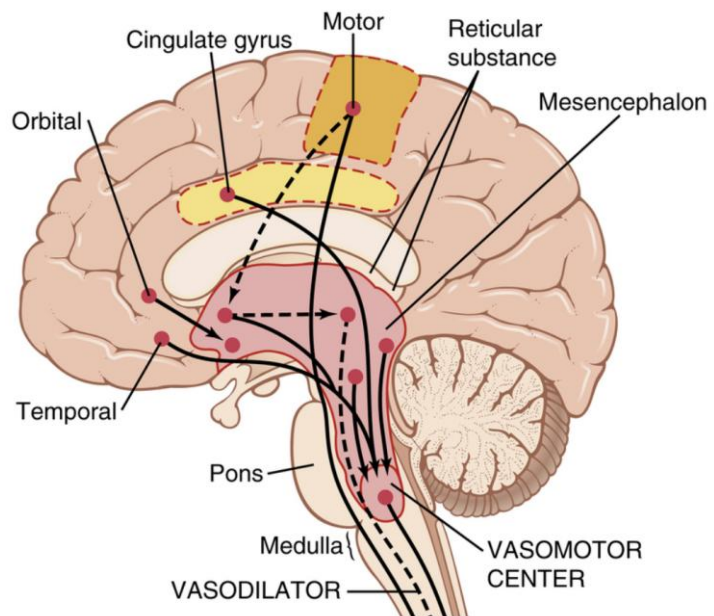


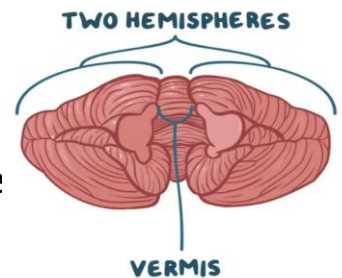
Figure for Illustration

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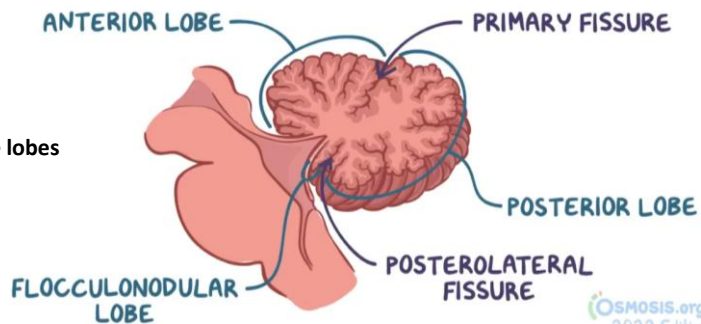
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Cerebellum “little brain”

- Located in the posterior cranial fossa behind the pons and medulla.
- Cerebellum consists of two hemispheres separate by a narrow, ridge named **vermis**.



Subdivided into three lobes



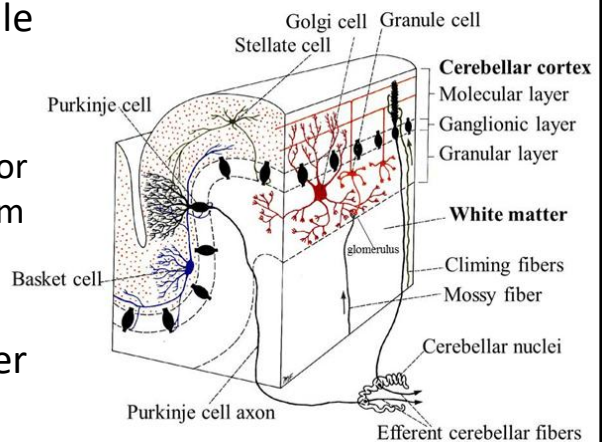
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Cerebellar cortex is made up of three layers: an outer molecular layer, middle Purkinje cell layer and inner granular layer.

- The Purkinje cell layer is responsible for the inhibitory output of the cerebellum using the inhibitory neurotransmitter gamma-aminobutyric acid (GABA).
- Underneath the cortex, white matter axons that carry information to and from the cerebellum.



Adopted from: R.S. Snell. Clinical and functional histology for medical students, Little, Brown and Company, 1984

Functions:

- Anterior lobe (spinocerebellum) is responsible for regulation of muscle tone.
- Posterior lobe (cerebrocerebellum) is responsible for planning fine motor movements & coordinate motor function.
- Flocculonodular lobe (vestibulocerebellum), responsible for the maintenance of posture and balance.
- Integrates sensory input with motor output to create coordinated motor function.
- Cerebellar dysfunction leads neurological disorders like hypotonia and ataxia

Ataxia

- Characterized by unstable & uncoordinated movement. Three main types with several etiologies: cerebellar, sensory and vestibular.
- **Cerebellar ataxia:**
 - Caused by cerebellar degeneration secondary to vascular occlusion, medications, infections, genetics (Ataxia Telangiectasia or Friedreich's Ataxia).
 - Friedreich's Ataxia is autosomal recessive caused by mutation of frataxin gene (ch 9). Characterized by degradation of cerebellum and spinal cord, leading to abnormal gait and others.
 - Ataxia Telangiectasia is caused by defective ATM gene (ch 11), leading to cerebellar atrophy, immune deficiency & others. Children have high risk of cancer and DNA is highly sensitive to ionizing radiation.

[Diagnostic dental radiographs not allowed! A child with ataxia telangiectasia - PubMed](#)

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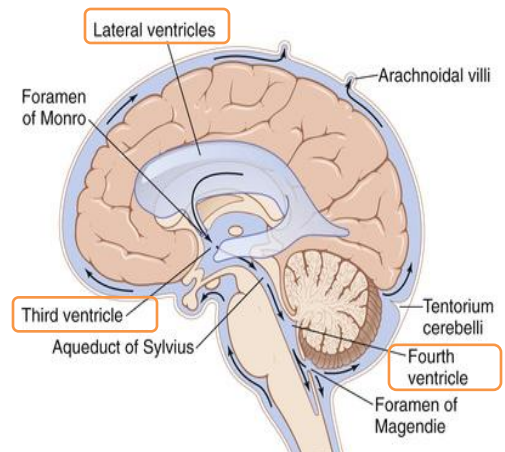
Brain ventricles

Four ventricles:

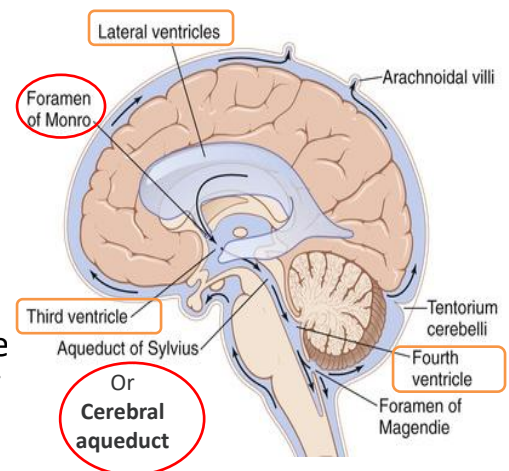
Two lateral ventricles, C-shaped chambers located within each cerebral hemisphere.

Third ventricle, narrow space located between the right and left diencephalon.

Fourth ventricle, posterior to the medulla and anterior to the cerebellum.



- The ventricular system is important for the production and circulation of the cerebrospinal fluid. The cerebrospinal fluid protects the brain by providing hydromechanical protection.
- Each lateral ventricle communicates via the **Foramen of Monro** with the third ventricle. The third ventricle communicates with the fourth ventricle via the **cerebral aqueduct**. From the fourth ventricle, the CSF goes to the subarachnoid space or to the central canal of the spinal cord to provide hydromechanical protection.



Hydrocephalus

Hydrocephalus refers to a diverse group of conditions that result from excessive cerebrospinal fluid production or impaired absorption of the cerebrospinal fluid or obstructions within the ventricular system. Hydrocephalus can be congenital or acquired

Accumulation of the cerebrospinal fluid in the lateral ventricles causes progressive ventricular dilatation, affecting brain growth and development.

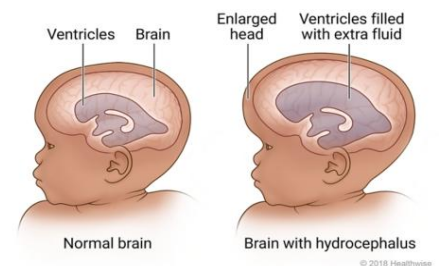
<https://pubmed.ncbi.nlm.nih.gov/20415804>

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➤ In children, hydrocephalus manifest as enlargement of the head because the skull bones are not completely fused yet.

- Symptoms: include headache, behavior changes, developmental delays, vomiting, and lethargy.
- Dental manifestations: changes in tooth eruption patterns, higher plaque accumulation and higher caries prevalence.



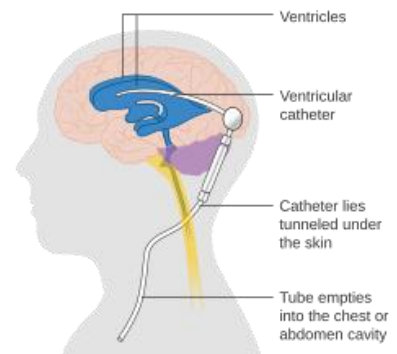
➤ In adults, the accumulation of cerebrospinal fluid compresses the brain tissue. Patients have declining cognitive function, headache, poor coordination,,,,,

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Treatment strategies: includes inserting an extracranial shunt into the ventricles to drain the excess cerebrospinal fluid into the abdominal cavity (VP) or heart atrium (VA).

Dental modifications: treatment plans are in coordination with neurology team. Include: chair positioning, avoid excessive force to the catheter (along the neck), referral for possible antibiotic prophylaxis (VA shunt), shorter appointment to enable tolerating care.,,



[Ventriculoperitoneal shunt](https://en.m.wikipedia.org)
en.m.wikipedia.org

Recommended resources



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